

CS 543 Web Services — Assignment 2

SCM (Smart Contact Manager): Services, Contracts & Schema

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Source model used: COMPLETE_SCM_SCHEMA.mwb. The model contains four tables: users, password_reset_token, contact, and social_link. The assignment brief requires 3–6 services, service contracts, a central operation, an ER schema, SQL, and validation against five service properties.

Important SCM adaptation: The brief names the central operation as *placeOrder* for CampusEats. SCM has no order domain in the supplied model, so the corresponding central SCM operation is **createContact**. No order tables or unrelated entities are invented.

1. System Capabilities

Capability	What the system does
Register account	Create a new SCM user account.
Authenticate user	Log in and establish an authenticated session.
Manage profile	View and update the user profile and account information.
Reset password	Request a password reset and apply a valid reset token.
Create contact	Add a new contact to the authenticated user's contact collection.
View contact	Retrieve the details of a specific contact.
Update contact	Modify an existing contact owned by the authenticated user.
Delete contact	Remove an existing contact owned by the authenticated user.
List contacts	Retrieve the authenticated user's contacts.
Manage favorites	Mark or unmark a contact as a favorite.
Manage social links	Add, update, view, and remove social links associated with a contact.

2. Service Design Benchmark

Service	Member	Owned data	Responsibilities
Identity Service	Aman Bali	users; password-reset credentials and lifecycle data	register, authenticate, manage profile, reset password
Contact Service	Bhupesh Kumar	contact	create, read, update, delete, list, favorite contacts
Social Link Service	Aman Bali	social_link	add, read, update, delete social links

Cross-service calls: Contact → Identity: *validateOwner(userId)*. Social Link → Contact: *getContact(contactId)*. These are service operations, not shared database ownership. The editable benchmark is `services.drawio`.

3. Service Contracts

Contracts expose operations and hide implementation details. They intentionally do not expose table names, SQL, or database-specific identifier formats.

Identity Service

Operation	Input	Output	Errors
register	email, userName, password, optional profile data	User account created; public profile summary	Validation failed; email already used; provider data invalid
authenticate	email/userName and password	Authenticated user summary and authentication result	Invalid credentials; account disabled
getProfile	Authenticated user context	Profile information	User not found; unauthorized
updateProfile	Authenticated user context and profile changes	Updated profile information	Invalid data; unauthorized
requestPasswordReset	Email or username	Reset request accepted	Account not found; request rejected
resetPassword	Reset token and new password	Password reset confirmation	Token invalid; token expired; weak password

Contact Service

Operation	Input	Output	Errors
createContact	Authenticated user context and contact details	Created contact representation	Invalid data; owner invalid; duplicate/conflict
getContact	Authenticated user context and contact reference	Contact representation	Not found; not owned by caller
listContacts	Authenticated user context and optional filters/page cursor	Contact collection page	Unauthorized; invalid cursor/filter
updateContact	Authenticated user context, contact reference, changed fields	Updated contact representation	Not found; not owned; validation failed
deleteContact	Authenticated user context and contact reference	Deletion confirmation	Not found; not owned
setFavorite	Authenticated user context, contact reference, favorite state	Updated favorite state	Not found; not owned

Social Link Service

Operation	Input	Output	Errors
addSocialLink	Authenticated user context, contact reference, title, link	Created social-link representation	Contact not found; unauthorized; invalid URL
getSocialLinks	Authenticated user context and contact reference	Social-link collection	Contact not found; unauthorized

Operation	Input	Output	Errors
updateSocialLink	Authenticated user context, social-link reference, changed fields	Updated social-link representation	Not found; unauthorized; invalid URL
deleteSocialLink	Authenticated user context and social-link reference	Deletion confirmation	Not found; unauthorized

4. Central Operation — SCM Equivalent of placeOrder

Operation: createContact

Purpose: Create a new contact for the authenticated SCM user. This is the domain-equivalent central write operation because the supplied SCM schema has no order/checkout data.

Aspect	Specification
Input	Authenticated caller context; name (required); optional address, description, email, Facebook/LinkedIn/website links, phone number, picture, and favorite state.
Output	A successful result containing the newly created contact representation and a reference that callers can use for subsequent operations.
Error cases	400 invalid contact data; 401 missing/invalid authentication; 403 caller cannot act for the requested owner; 409 conflicting contact state; 500 unexpected service failure.
Internal details hidden	The caller does not know how the service validates fields, persists the contact, generates the database key, commits the transaction, or maps persistence errors to the public response.
Cross-service behavior	Before creation, Contact Service may call Identity Service using validateOwner(userId) to confirm that the authenticated owner is valid. The Contact Service remains the owner of contact data.

5. Database Schema

The supplied Workbench model has four tables. Ownership is assigned to exactly one service: Identity owns users and password_reset_token; Contact owns contact; Social Link owns social_link.

Service-boundary adjustment: The source Workbench model contains relational links from contact to users and social_link to contact. In the service benchmark SQL, those cross-service relationships are retained as opaque references rather than database foreign keys. The only database FK is the internal Identity relationship password_reset_token → users. This prevents one service from requiring another service's database to be available for local writes.

Owned table	Columns / keys
Identity — users	user_id BIGINT PK; about VARCHAR(255); email VARCHAR(255) NOT NULL; email_verified TINYINT; enabled TINYINT; user_name VARCHAR(100) NOT NULL; password VARCHAR(255); phone_number VARCHAR(255); phone_verified TINYINT; profile_pic VARCHAR(255); provider ENUM(FACEBOOK,GITHUB,GOOGLE,SELF); provider_user_id VARCHAR(255).
Identity — password_reset_token	token VARCHAR(255) PK; expiry_date DATETIME NOT NULL; user_user_id BIGINT FK → users.user_id.
Contact — contact	id INT PK; address VARCHAR(255); description VARCHAR(255); email VARCHAR(255); facebook_link VARCHAR(255); favorite TINYINT; linkedin_link VARCHAR(255); name VARCHAR(255) NOT NULL; phone_number VARCHAR(255); picture VARCHAR(255); website_link VARCHAR(255); user_user_id BIGINT opaque Identity reference.
Social Link — social_link	id INT PK; link VARCHAR(255) NOT NULL; title VARCHAR(255) NOT NULL; contact_id INT opaque Contact reference.

Equivalent CREATE TABLE statements are provided separately in schema.sql.

6. Service Property Validation

Service	Property	Result	Evidence / fix
Identity Service	Reachable	Pass	REST endpoint is reachable by clients.
Identity Service	Self-contained	Pass	Owns user and password-reset data; local persistence does not require Contact or Social Link DB access.
Identity Service	Has a contract	Pass	Six public operations are defined above.
Identity Service	Independent	Pass	Can evolve its implementation without exposing database details.
Identity Service	Loosely coupled	Pass	Other services call operations rather than reading its database.
Contact Service	Reachable	Pass	REST endpoints expose contact operations.
Contact Service	Self-contained	Pass	Owns contact data and local contact behavior.
Contact Service	Has a contract	Pass	Six public operations are defined above.
Contact Service	Independent	Pass	Uses Identity only through validateOwner; no shared DB ownership.
Contact Service	Loosely coupled	Pass	One narrow validation call to Identity; no direct DB access.
Social Link Service	Reachable	Pass	REST endpoints expose social-link operations.
Social Link Service	Self-contained	Pass	Owns social_link data and its local operations.
Social Link Service	Has a contract	Pass	Four public operations are defined above.
Social Link Service	Independent	Pass	Contact is accessed through getContact, not shared storage.
Social Link Service	Loosely coupled	Pass	One narrow dependency on Contact Service.

Overall result: The benchmark satisfies the five service properties for all three services. The main architectural safeguard is that cross-service relationships are represented as service-level references and calls, not shared database ownership.

Deliverables Checklist

- ✓ design.pdf — capabilities, service ownership, contracts, central operation, validation
- ✓ services.drawio — editable service benchmark
- ✓ services.png — exported service benchmark
- ✓ schema.drawio — editable ER/schema diagram
- ✓ schema.png — exported schema diagram
- ✓ schema.sql — CREATE TABLE implementation sketch